Unit: Unit 0: Foundations of Algebra **Name:** ____________
Topic: Inequalities **Date:** ________

Quiz (Carolina Reaper)

Q1. A cargo ship travels 120 miles in t hours. If the speed is at least 20 mph, what inequality represents this situation?
(A) $t \geq 6$
(B) $t \leq 6$
(C) $t > 6$
(D) $t < 6$
(E) $t = 6$

Q2. The cost C of shipping a package is given by $C = 5x + 10$, where x is the weight in pounds. If the cost is at most

$$25$$

, what is the inequality for x?
(A) $x \leq 3$
(B) $x \geq 3$
(C) $x < 3$
(D) $x > 3$
(E) $x = 3$

Q3. A traffic light has a red phase that lasts for t seconds. If t is between 20 and 40 seconds, the inequality for t is
(A) $20 < t < 40$
(B) $20 \leq t \leq 40$
(C) $t > 20$
(D) $t < 40$
(E) $t = 20$

Q4. The number of boxes n that can be delivered by a truck is given by $n = 10t - 5$, where t is the time in hours. If the number of boxes is at least 30, what inequality represents this situation?
(A) $t \geq 3.5$
(B) $t \leq 3.5$
(C) $t > 3.5$
(D) $t < 3.5$
(E) $t = 3.5$

Q5. A company offers a discount on shipping for orders over 50. If x represents the cost of the order, the inequality for orders that qualify for the discount is
(A) $x > 50$
(B) $x \geq 50$
(C) $x < 50$
(D) $x \leq 50$
(E) $x = 50$

Q6. A delivery person can carry at most 25 packages. If x represents the number of packages, the inequality for x is
(A) $x \leq 25$
(B) $x \geq 25$
(C) $x < 25$
(D) $x > 25$
(E) $x = 25$

Q7. The time t it takes for a package to be delivered is given by $t = \frac{x}{10} + 2$, where x is the distance in miles. If the delivery time is at most 5 hours, what inequality represents this situation?
(A) $x \leq 30$
(B) $x \geq 30$
(C) $x < 30$
(D) $x > 30$
(E) $x = 30$

Q8. A truck can travel at most 400 miles in a day. If x represents the distance traveled, the inequality for x is
(A) $x \leq 400$
(B) $x \geq 400$
(C) $x < 400$
(D) $x > 400$
(E) $x = 400$

Open Ended Questions (FRQ)

Q9. A package delivery company guarantees that packages will be delivered within t hours, where $t = \frac{x}{5} + 1$, and x is the distance in miles. Graph the inequality $t \leq 4$ on a number line and write the solution in interval notation.

Q10. A cargo ship travels from port A to port B at a speed of s miles per hour. The time t it takes for the ship to travel from A to B is given by $t = \frac{240}{s}$. If the time t is between 4 and 12 hours, write an inequality for s and graph the solution on a number line.

Chef's Tips

- Make sure students show all work and explain their reasoning.
- Emphasize the importance of checking the units of the variables.
- Encourage students to use different methods to solve the problems, such as graphing and algebraic manipulation.